



Total Island = 0

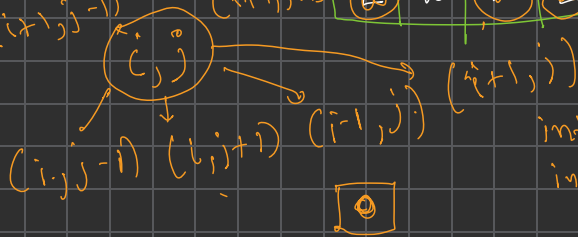
DFS

0	1	1	1	1
0	1	w	w	w
w	1	w	w	1
1	w	w	1	1
w	w	w	w	w
1	w	1	1	w

L = Land
w = Water

Total no of Island?

$(i-1, j-1)$ $(i-1, j+1)$
 $(i+1, j-1)$ $(i+1, j+1)$



$int row[4] = \{0, 0, -1, 1\}$
 $int col[4] = \{1, -1, 0, 0\}$

$int row[8] = \{0, 0, -1, 1, -1, -1, 1, 1\}$
 $int col[8] = \{-1, 1, 0, 0, -1, 1, -1, 1\}$

For $(i=0; i < 8; i++)$ DFS $(grid, 0, 0)$
 $\{ (i+row[k], j+col[k]) \}$ $(grid, 0, 0)$
 $grid[i][j]$

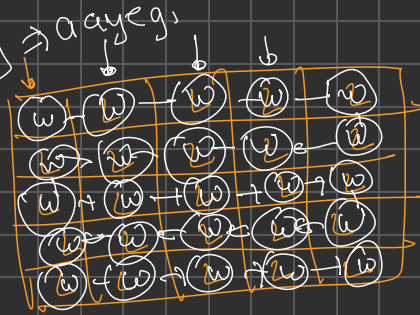
Total 4

0	1	1	w	w	w
1	w	1	w	w	w
2	1	w	w	w	w
3	w	w	w	w	w
4	1	w	1	1	w

$n \times m$ ✓
 $O(nm)$

$\text{int row}[8] = \{0, 0, 1, 1, 1, -1, -1, -1\}$
 $\text{int col}[8] = \{0, -1, 0, 1, -1, 0, 1, -1\}$

$O(nm)$
 $n \times m$
 $n \times m$



space complexity

$n \times m$
 $O(n \times m)$

w	L	L	L	w
w	w	L	w	w
w	w	w	w	w
L	L	w	w	L
w	L	w	w	L

$O(L, L, L)$

	0	1	2	3	4	5
0	x	x	x	x	0	x
1	x	x	x	0	0	x
2	x	x	x	x	0	x
3	x	x	x	x	0	x
4	x	x	x	x	x	x
5	x	x	x	x	0	0

Don't dep
14 m

yahi

replace

$(1,1)$ $(4,1)$ $(2,2)$
 $(2,1)$ $(3,2)$
 $(3,1)$

